Periodic Trends

Elements are arranged on the periodic table based on their physical and chemical properties.

It was **Dmitri Mendeleev** (1834-1907) who first organized a table listing the elements in order of atomic mass in vertical columns. Each of the columns ended when the chemical properties of the elements started to repeat themselves.

What are the physical & chemical property trends in the table?

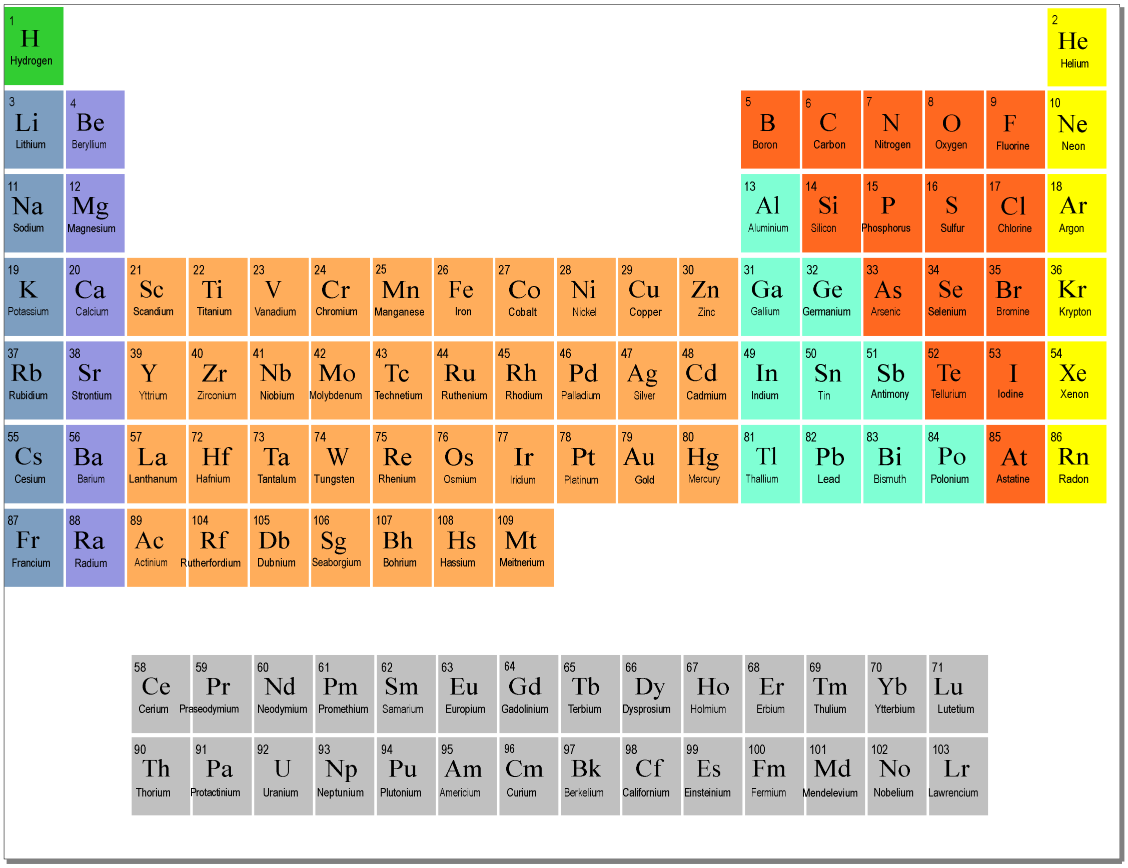
1) Metals, non-metals, metaolloids
   - metals are good conductors of heat and electricity, have high densities, high melting points, high boiling points, they are ductile, lustrous, etc.

2) The period # (row) is the number of energy levels present in the atom

3) The group # (column) is the number of valence electrons

4) Atoms are listed in increasing atomic mass

5) Atoms are listed in increasing atomic number



6) The *__Atomic Radius__* is the distance from the nucleus to the valence shell of an atom. AR increases down and to the left.

7) *__Ionization Energy__* is the energy necessary to remove an electron from a neutral atom. IE increases up and to the right.

8) The *__Electron Affinity__* of an element is the energy given off when a neutral atom gains an extra electron. EA increases up and to the right.

9) *__Electronegativity__* is a measure of the ability of an atom to attract electrons in the context of a chemical bond. The electronegativity is a relative value between 0 and 4. Electronegativity increases up and to the right.

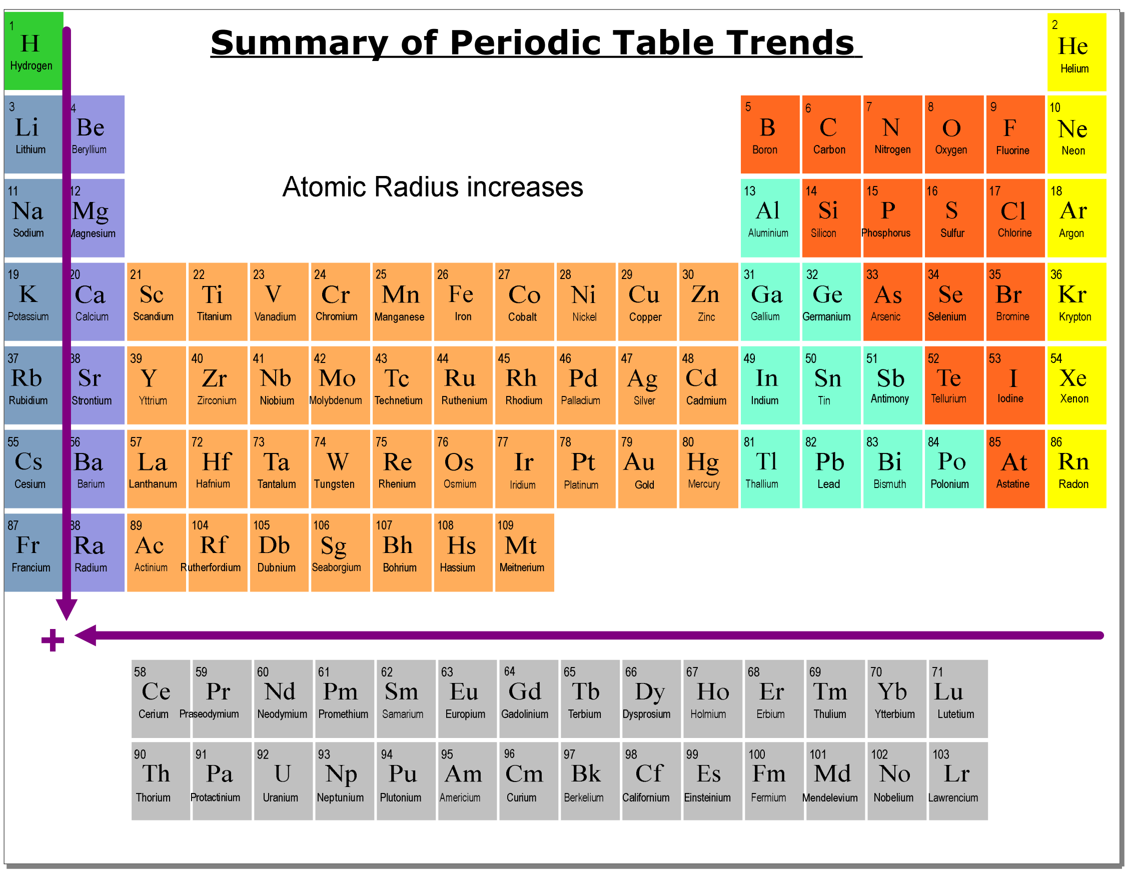
Summary of Periodic Table Trends

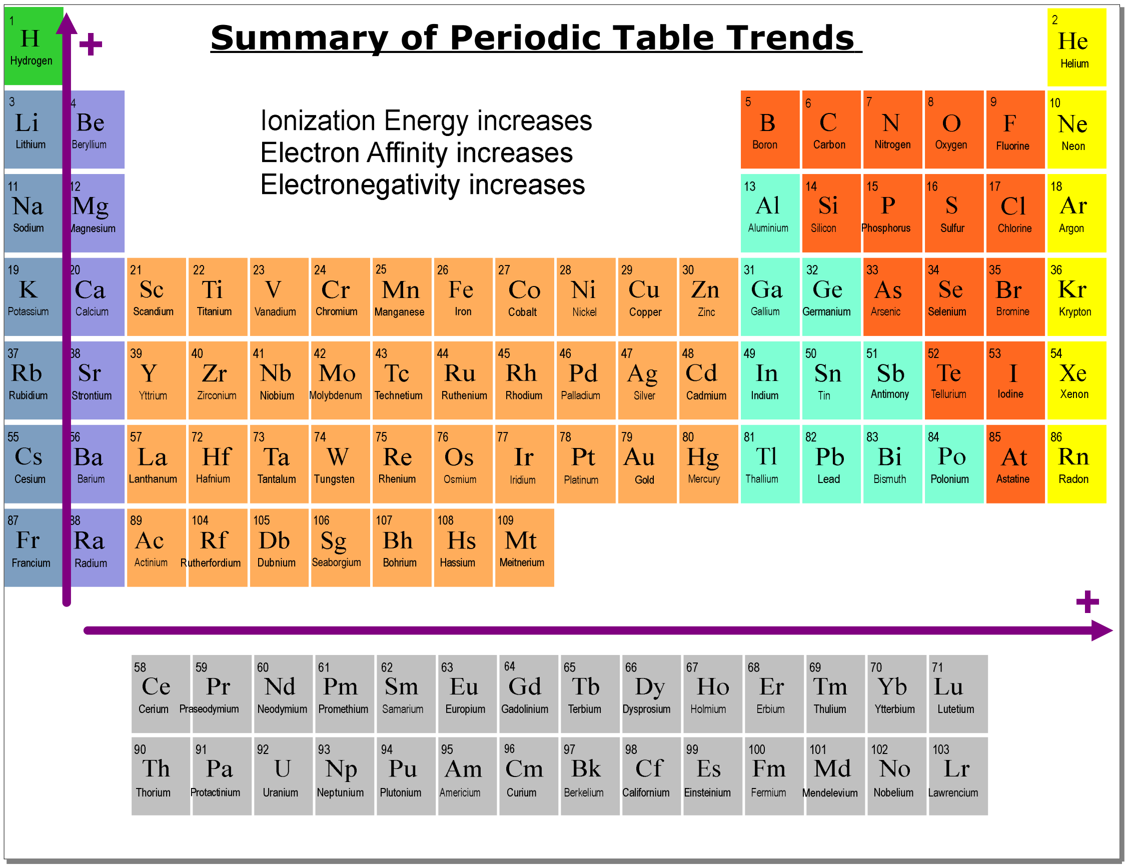
Moving Left --> Right

- Atomic Radius Decreases
- Ionization Energy Increases
- Electron Affinity Increases
- Electronegativity Increases

Moving Top --> Bottom

- Atomic Radius Increases
- Ionization Energy Decreases
- Electron Affinity Decreases
- Electronegativity Decreases





Read: pg 31-38

Periodic Trend Assignment

Due: